

Pratham Raj

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pratham2411.online | [LeetCode: prathamraj2411](https://leetcode.com/prathamraj2411) | [Codeforces: prathamraj2411](https://codeforces.com/prathamraj2411)

EDUCATION

National Institute of Technology Patna

B.Tech in Electrical Engineering

Patna, India

Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript (ES6+), TypeScript, HTML5, CSS3
Backend & APIs: Node.js, Express.js, RESTful APIs, JWT Authentication, OAuth 2.0, Socket/Network Programming
Frontend: React.js, Zustand, Redux, Tailwind CSS, Vite, Responsive Design, PWA
Databases & Cloud: MongoDB, MySQL, Redis, Cloudinary, Vercel, Render, Mongoose ODM
Tools & Practices: Git/GitHub, Postman, Linux/Unix, CMake, Docker (basic), Agile, CI/CD, Code Review
Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Multi-threading, Scalability, Deep Packet Inspection

EXPERIENCE

Web Developer Intern

May 2026 – July 2026

[InternPe \(Certificate\)](#) | [GitHub](#) | [Live](#)

Remote

- Engineered an Interactive Question Sheet SPA using **React.js** and **Zustand** with a Topic → Subtopic → Question hierarchy, drag-and-drop reordering (@dnd-kit), and real-time progress tracking.
- Designed accessible, responsive UI components with **Tailwind CSS**; deployed on **Vercel** with `localStorage` persistence and **Firebase** authentication, achieving sub-second page load times and cross-device sync.
- Practiced **agile** workflows with **Git** version control, iterative feature delivery, and cross-browser compatibility testing across 3 major browsers.

PROJECTS

VectorForge – C++ Vector Database & RAG Engine | C++17, HNSW, KD-Tree, cpp-httplib, Ollama | GitHub Jan 2026 – Present

- Built a **vector database from scratch** in C++ implementing three interchangeable K-NN algorithms – Brute Force $O(N)$, KD-Tree, and **HNSW** ($M=16$, $ef_{construction}=200$) with Cosine, Euclidean, and Manhattan distance metrics, demonstrating how Pinecone, Weaviate, and Chroma work under the hood.
- Implemented a **RAG pipeline**: text → Ollama (nomic-embed-text, **768D embeddings**) → HNSW semantic search → **Llama 3.2** generation, with 250-word overlapping chunking across multi-document stores.
- Exposed a **REST API** via `cpp-httplib` (insert, delete, search, benchmark, hnsw-info endpoints) with a live **2D PCA scatter plot** visualizing semantic clusters of 20 pre-loaded vectors across 4 categories.

NIT Patna Market | React.js, Node.js, MongoDB, Cloudinary, JWT | GitHub | Live

Aug 2025 – Present

- Led full-stack **MERN** development with **JWT authentication**, **role-based access control** (guest/user/seller/admin), and OTP-verified `@nitp.ac.in` signup via Resend API, securely serving **1,000+ university students**.
- Built dual-mode **image pipeline** (Cloudinary CDN + local-disk fallback) for up to 8 images/listing, reducing image load times by **30%**; buyer-seller **HTTP polling chat** with read receipts and cron-scheduled email digest.
- Deployed as a **PWA** on Vercel + Render with optimistic UI enabling **500+ daily transactions**; indexed **MongoDB** queries with Mongoose ODM achieving sub-100ms response times.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

Amazon HackOn 6.0: Ranked **Top 75 coders** globally out of **75,000+** participants.

Competitive Programming: Knight Badge (2000+ peak rating) on LeetCode | **3-Star** (1650+) on CodeChef | **Pupil** (1250+) on Codeforces | **2000+** problems solved.

Hackathon Champion: Secured **1st place** at Hackslash College Hackathon – rapid prototyping and algorithmic optimization under strict time constraints.

Industry Recognition: Shortlisted for **Digital Specialist Engineer (DSE)** at Infosys via top-percentile technical assessments. Ranked **Top 10** institution-wide on Codolio leaderboard at NIT Patna.

LEADERSHIP & VOLUNTEERING

Campus Ambassador, HackerRank & GfG: Facilitated DSA workshops, mentoring peers in competitive programming and algorithmic thinking.

Educational Volunteer, NSS: Instructed **500+** students, spearheading community-driven educational outreach initiatives.

Technical Contributor, GDSC: Architected open-source software solutions, collaborating to foster campus developer communities.